AMBIKA DAS

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EDUCATION

Bachelor's Degree, Computer Science Sep 2021 – July 2024
Maulana Abul Kalam Azad University of Technology, WB, India *CGPA: 8.51/ 10.00*

- Worked as Field Secretary of the college committee and helped organize cultural and technical events.
- Founded and worked in "Techno Endeavors", a club that supported underprivileged students.
- Completed two Full Stack Java projects using React.js, Spring Boot, JSP, MySQL, and Oracle.

SKILLS

Programming Languages:	Python3, Java, C Programming, UNIX and Shell Programming
Machine Learning & Deep Learning:	Machine Learning Using Python, Federated Learning.
Libraries & Frameworks:	PyTorch, NumPy, Pandas, Matplotlib, Seaborn, Stable Diffusion, Hugging Face, Git & GitHub.
Development Skills:	HTML5, CSS3, MySQL, Oracle 10g
Soft Skills:	Problem Solving, Research, Independent Development, Team Collaboration, Communication.

EXPERIENCES

Research Fellow, *Indian Institute of Technology, Patna, Bihar, India* Aug 2025 – Nov 2025

- Worked on Secure **Blockchain-based Federated Learning (SBFL)** for decentralized AI systems that remain safe even during attacks.
- Running experiments on MNIST, CIFAR-10, and CIFAR-100 datasets under IID and non-IID settings.
- Supporting research work for conference and journal publications.

Research Intern, *India Institute of Technology, Patna, Bihar, India* Nov 2024 – July 2025

- Worked on secure Federated Learning systems.
- Created and tested attacks like **Backdoor** and **Model Poisoning**, and developed defense methods such as **Bucketing-Filtering** and GAN-based protection.
- Improved **coding**, **research**, and analytical skills through practical experiments.

PROJECTS

Blockchain-based Fully Decentralized Federated Learning Aug 2025 – Nov 2025

- Implemented blockchain-based decentralized Federated Learning frameworks.
- Used consensus methods and digital signatures to make training secure.
- Tested the system against attacks like label flipping and malicious clients.
- Evaluated accuracy and security on MNIST and CIFAR-10 datasets.

Federated Learning Model in Attacks and Byzantine-Robust Aggregators Nov 2024 – July 2025

- Working on a **Federated Learning project** during my internship at **IIT Patna**, to build models that stay reliable even if attacked.
- Testing different types of attacks like **Backdoor**, **Model Poisoning**, and **Krum**.
- Developing ways to protect the model using methods like **Bucketing-Filtering** and **Krum-**based filtering.

Crop Prediction using Machine Learning Aug 2023 – Sep 2023

- Build a crop prediction system using **2000 data** records.
- Cleaned and processed data using **Pandas**.
- Applied **Logistic Regression** to predict crop outcomes.
- Used **Matplotlib** to create visual reports and heatmaps.

Project Link - https://github.com/ambikad04/Machine-Learning-Python-Crop-Prediction-project

RESEARCH PAPER

Empowering Agricultural Sustainability: Integrating Nanosensors and Machine Learning for Informed Crop Management in India **(**Publish on March 2025)

Publisher: APPLE ACADEMIC PRESS

- This research paper shows how machine learning can improve crop prediction and farming decisions using **Logistic Regression.**

Link: **https://www.appleacademicpress.com/nanotechnology-and-beyond-the-synergy-of-materials- and-artificial-intelligence-/9781779643025**